

Concept 1 | Arithmetic Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Test

$$a_{n+1} - a_n = d$$

Worked Solutions

Arithmetic Sequences

Q001 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 2n - 3$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -1$, $a_2 = 1$, $a_3 = 3$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -1, a_2 = 1, a_3 = 3$$

Q002 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 7n$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -6$, $a_2 = -10$, $a_3 = -12$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -6, a_2 = -10, a_3 = -12$$

Worked Solutions

Arithmetic Sequences

Q003 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 5n - 8$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -3$, $a_2 = 2$, $a_3 = 7$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -3, a_2 = 2, a_3 = 7$$

Q004 Written

When the general term of the sequence $\{a_n\}$ is $a_n = -3n + 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -2$, $a_2 = -5$, $a_3 = -8$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -2, a_2 = -5, a_3 = -8$$

Worked Solutions

Arithmetic Sequences

Q005 Written

When the general term of the sequence $\{a_n\}$ is $a_n = -2n - 4$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -6$, $a_2 = -8$, $a_3 = -10$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -6, a_2 = -8, a_3 = -10$$

Q006 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 0$, $a_2 = 3$, $a_3 = 8$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 0, a_2 = 3, a_3 = 8$$

Worked Solutions

Arithmetic Sequences

Q007 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + n + 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 7$, $a_3 = 13$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 3, a_2 = 7, a_3 = 13$$

Q008 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + 2n + 4$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 7$, $a_2 = 12$, $a_3 = 19$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 7, a_2 = 12, a_3 = 19$$

Worked Solutions

Arithmetic Sequences

Q009 MCQ

For $a_n = 3n - 5$, the first three terms (a_1, a_2, a_3) are:**Solution**

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -2$, $a_2 = 1$, $a_3 = 4$.
3. Check each value by putting its index back into the formula.

Correct option: B**Final answer** $(-2, 1, 4)$

Q010 MCQ

For $a_n = 4n + 7$, the first three terms (a_1, a_2, a_3) are:**Solution**

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 11$, $a_2 = 15$, $a_3 = 19$.
3. Check each value by putting its index back into the formula.

Correct option: C**Final answer** $(11, 15, 19)$

Worked Solutions

Arithmetic Sequences

Q011 MCQ

For $a_n = n^2 - 4n + 6$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 2$, $a_3 = 3$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

(3, 2, 3)

Q012 MCQ

For $a_n = 2n^2 + 3$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 5$, $a_2 = 11$, $a_3 = 21$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

(5, 11, 21)

Worked Solutions

Arithmetic Sequences

Q013 MCQ

For $a_n = n^2 + 5n - 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 4$, $a_2 = 10$, $a_3 = 16$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

$(4, 10, 16)$

Q014 MCQ

For $a_n = 5 - 2n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 1$, $a_3 = -1$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$(3, 1, -1)$

Worked Solutions

Arithmetic Sequences

Q015 MCQ

For $a_n = 2n^2 - 7n + 9$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 4$, $a_2 = 3$, $a_3 = 6$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

(4, 3, 6)

Q016 MCQ

For $a_n = n^2 - 3n - 1$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -3$, $a_2 = -3$, $a_3 = -1$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

(-3, -3, -1)

Worked Solutions

Arithmetic Sequences

Q017 MCQ

For $a_n = 3n^2 - 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = 10$, $a_3 = 25$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

(1, 10, 25)

Q018 MCQ

For $a_n = n^2 + 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 6$, $a_3 = 11$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

(3, 6, 11)

Worked Solutions

Arithmetic Sequences

Q019 MCQ

For $a_n = 7n - 10$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -3$, $a_2 = 4$, $a_3 = 11$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

$(-3, 4, 11)$

Q020 MCQ

For $a_n = 2n^2 + n - 1$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 9$, $a_3 = 20$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

$(2, 9, 20)$

Worked Solutions

Arithmetic Sequences

Q021 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 2 \times 3^n$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 6$, $a_2 = 18$, $a_3 = 54$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 6, a_2 = 18, a_3 = 54$$

Q022 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 3^n - 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 8$, $a_3 = 26$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 2, a_2 = 8, a_3 = 26$$

Worked Solutions

Arithmetic Sequences

Q023 Written

When the general term of the sequence $\{a_n\}$ is $a_n = (-3)^n$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -3$, $a_2 = 9$, $a_3 = -27$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -3, a_2 = 9, a_3 = -27$$

Q024 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 3 \times (-2)^n$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -6$, $a_2 = 12$, $a_3 = -24$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -6, a_2 = 12, a_3 = -24$$

Worked Solutions

Arithmetic Sequences

Q025 MCQ

For $a_n = 2^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 4$, $a_3 = 8$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

$(2, 4, 8)$

Q026 MCQ

For $a_n = (-2)^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -2$, $a_2 = 4$, $a_3 = -8$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

$(-2, 4, -8)$

Worked Solutions

Arithmetic Sequences

Q027 MCQ

For $a_n = 2 \times 3^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 6$, $a_2 = 18$, $a_3 = 54$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

(6, 18, 54)

Q028 MCQ

For $a_n = 5 \times (-3)^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -15$, $a_2 = 45$, $a_3 = -135$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

(-15, 45, -135)

Worked Solutions

Arithmetic Sequences

Q029 MCQ

For $a_n = 4^n - 1$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 15$, $a_3 = 63$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

(3, 15, 63)

Q030 MCQ

For $a_n = 2 \times 3^{n-1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 6$, $a_3 = 18$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

(2, 6, 18)

Worked Solutions

Arithmetic Sequences

Q031 MCQ

For $a_n = (-1)^{n-1}5^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 5$, $a_2 = -25$, $a_3 = 125$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

(5, -25, 125)

Q032 MCQ

For $a_n = (-3)^n + 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -1$, $a_2 = 11$, $a_3 = -25$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

(-1, 11, -25)

Worked Solutions

Arithmetic Sequences

Q033 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{n+1}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{2}$, $a_2 = \frac{2}{3}$, $a_3 = \frac{3}{4}$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = \frac{1}{2}, a_2 = \frac{2}{3}, a_3 = \frac{3}{4}$$

Q034 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{2n+1}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{3}$, $a_2 = \frac{2}{5}$, $a_3 = \frac{3}{7}$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = \frac{1}{3}, a_2 = \frac{2}{5}, a_3 = \frac{3}{7}$$

Worked Solutions

Arithmetic Sequences

Q035 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{1}{3n-2}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = \frac{1}{4}$, $a_3 = \frac{1}{7}$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 1, a_2 = \frac{1}{4}, a_3 = \frac{1}{7}$$

Q036 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{2n}{n+3}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{2}$, $a_2 = \frac{4}{5}$, $a_3 = 1$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = \frac{1}{2}, a_2 = \frac{4}{5}, a_3 = 1$$

Worked Solutions

Arithmetic Sequences

Q037 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n+1}{2n-1}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 1$, $a_3 = \frac{4}{5}$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 2, a_2 = 1, a_3 = \frac{4}{5}$$

Q038 MCQ

For $a_n = \frac{n+1}{n+2}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{2}{3}$, $a_2 = \frac{3}{4}$, $a_3 = \frac{4}{5}$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$$\left(\frac{2}{3}, \frac{3}{4}, \frac{4}{5}\right)$$

Worked Solutions

Arithmetic Sequences

Q039 MCQ

For $a_n = \frac{2n-1}{3n+1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{4}$, $a_2 = \frac{3}{7}$, $a_3 = \frac{1}{2}$.
3. Check each value by putting its index back into the formula.

Correct option: D**Final answer**

$$\left(\frac{1}{4}, \frac{3}{7}, \frac{1}{2}\right)$$

Q040 MCQ

For $a_n = \frac{n^2+1}{n+1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = \frac{5}{3}$, $a_3 = \frac{5}{2}$.
3. Check each value by putting its index back into the formula.

Correct option: A**Final answer**

$$\left(1, \frac{5}{3}, \frac{5}{2}\right)$$

Worked Solutions

Arithmetic Sequences

Q041 MCQ

For $a_n = \frac{3n}{2n-1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 2$, $a_3 = \frac{9}{5}$.
3. Check each value by putting its index back into the formula.

Correct option: B**Final answer** $(3, 2, \frac{9}{5})$

Q042 MCQ

For $a_n = \frac{2n+3}{n+4}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = \frac{7}{6}$, $a_3 = \frac{9}{7}$.
3. Check each value by putting its index back into the formula.

Correct option: C**Final answer** $(1, \frac{7}{6}, \frac{9}{7})$

Worked Solutions

Arithmetic Sequences

Q043 MCQ

For $a_n = \frac{n}{3n-2}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = \frac{1}{4}$, $a_3 = \frac{3}{7}$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

$$(1, \frac{1}{4}, \frac{3}{7})$$

Q044 Written

For the sequence $\{a_n\} : 0, 1, 4, 9, 16, \dots$, deduce the general term.

Solution

1. The terms are (term number $- 1$)², so $a_n = (n - 1)^2$.
2. Checking $n = 1, 2, 3$ gives 0,1,4.

Final answer

$$a_n = (n - 1)^2$$

Worked Solutions

Arithmetic Sequences

Q045 Written

For the sequence $\{a_n\} : 0, 1, 8, 27, 64, \dots$, deduce the general term.

Solution

1. The terms are successive cubes starting with 0: $(n - 1)^3$.
2. The formula gives 0,1,8,27,64 for $n = 1, 2, 3, 4, 5$.

Final answer

$$a_n = (n - 1)^3$$

Q046 Written

For the sequence $\{a_n\} : 1, 4, 9, 16, 25, \dots$, deduce the general term.

Solution

1. These are the squares $1^2, 2^2, 3^2, \dots$, so $a_n = n^2$.

Final answer

$$a_n = n^2$$

Worked Solutions

Arithmetic Sequences

Q047 **Written**

For the sequence $\{a_n\} : 7^2, 8^2, 9^2, 10^2, 11^2, \dots$, deduce the general term.

Solution

1. The bases are $n + 6$, so $a_n = (n + 6)^2$.

Final answer

$$a_n = (n + 6)^2$$

Q048 **MCQ**

Which formula gives the general term of $\{a_n\} : 1, 8, 27, 64, \dots$?:

Solution

1. Separate the visible pattern: the rule is $a_n = n^3$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: B

Final answer

$$n^3$$

Worked Solutions

Arithmetic Sequences

Q049 MCQ

Which formula gives the general term of $\{a_n\}$: 9, 16, 25, 36, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (n + 2)^2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: C**Final answer**

$$(n + 2)^2$$

Q050 MCQ

Which formula gives the general term of $\{a_n\}$: -1, 0, 1, 8, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (n - 2)^3$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: D**Final answer**

$$(n - 2)^3$$

Worked Solutions

Arithmetic Sequences

Q051 MCQ

Which formula gives the general term of $\{a_n\}$: 36, 49, 64, 81, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (n + 5)^2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: A**Final answer**

$$(n + 5)^2$$

Q052 MCQ

Which formula gives the general term of $\{a_n\}$: 1, 9, 25, 49, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (2n - 1)^2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: B**Final answer**

$$(2n - 1)^2$$

Worked Solutions

Arithmetic Sequences

Q053 MCQ

Which formula gives the general term of $\{a_n\}$: 0, 1, 16, 81, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (n - 1)^4$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: C**Final answer**

$$(n - 1)^4$$

Q054 Written

For the sequence $\{a_n\}$: 2, 5, 8, 11, 14, ..., deduce the general term.**Solution**

1. The common increase is 3 and the first term is 2, so $a_n = 2 + 3(n - 1) = 3n - 1$.

Final answer

$$a_n = 3n - 1$$

Worked Solutions

Arithmetic Sequences

Q055 Written

For the sequence $\{a_n\} : 0, 2, 4, 6, 8, \dots$, deduce the general term.

Solution

1. The common difference is 2 and $a_1=0$, so $a_n=2(n-1)=2n-2$.

Final answer

$$a_n = 2n - 2$$

Q056 Written

For the sequence $\{a_n\} : 0, 3, 6, 9, 12, \dots$, deduce the general term.

Solution

1. The common difference is 3 and $a_1=0$, so $a_n=3(n-1)=3n-3$.

Final answer

$$a_n = 3n - 3$$

Worked Solutions

Arithmetic Sequences

Q057 **Written****For the sequence $\{a_n\}$: 4, 9, 14, 19, 24, ..., deduce the general term.****Solution**

1. The common difference is 5 and $a_1=4$, so $a_n=4+5(n-1)=5n-1$.

Final answer

$$a_n = 5n - 1$$

Q058 **Written****For the sequence $\{a_n\}$: 6, 13, 20, 27, 34, ..., deduce the general term.****Solution**

1. The common difference is 7 and $a_1=6$, so $a_n=6+7(n-1)=7n-1$.

Final answer

$$a_n = 7n - 1$$

Worked Solutions

Arithmetic Sequences

Q059 MCQ

Which formula gives the general term of $\{a_n\}$: 7, 12, 17, 22, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = 5n + 2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: A**Final answer**

$$5n + 2$$

Q060 MCQ

Which formula gives the general term of $\{a_n\}$: -4, -1, 2, 5, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = 3n - 7$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: B**Final answer**

$$3n - 7$$

Worked Solutions

Arithmetic Sequences

Q061 MCQ

Which formula gives the general term of $\{a_n\}$: 10, 6, 2, -2, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = 14 - 4n$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: C**Final answer**

$$14 - 4n$$

Q062 MCQ

Which formula gives the general term of $\{a_n\}$: $\frac{1}{2}$, 1, $\frac{3}{2}$, 2, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = \frac{n}{2}$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: D**Final answer**

$$\frac{n}{2}$$

Worked Solutions

Arithmetic Sequences

Q063 Written

For the sequence $\{a_n\} : 2, -4, 6, -8, 10, \dots$, deduce the general term.

Solution

1. The magnitude is $2n$ and the first sign is positive with alternation, so the sign factor is $(-1)^{n-1}$.
2. Therefore $a_n = (-1)^{n-1} 2n$.

Final answer

$$a_n = (-1)^{n-1} 2n$$

Q064 Written

For the sequence $\{a_n\} : -3, 6, -9, 12, -15, \dots$, deduce the general term.

Solution

1. The magnitude is $3n$; the first term is negative and signs alternate, giving $(-1)^n$.
2. Therefore $a_n = (-1)^n 3n$.

Final answer

$$a_n = (-1)^n 3n$$

Worked Solutions

Arithmetic Sequences

Q065 Written

For the sequence $\{a_n\} : -5, 10, -15, 20, -25, \dots$, deduce the general term.

Solution

1. The magnitude is $5n$ and the first sign is negative, so $a_n = (-1)^n 5n$.

Final answer

$$a_n = (-1)^n 5n$$

Q066 Written

For the sequence $\{a_n\} : -1, 4, -7, 10, -13, \dots$, deduce the general term.

Solution

1. The magnitudes 1, 4, 7, 10, 13 are $3n - 2$; the first sign is negative, so $a_n = (-1)^n (3n - 2)$.

Final answer

$$a_n = (-1)^n (3n - 2)$$

Worked Solutions

Arithmetic Sequences

Q067 Written

For the sequence $\{a_n\} : 1, \frac{2}{3}, \frac{3}{5}, \frac{4}{7}, \frac{5}{9}, \dots$, deduce the general term.

Solution

1. The numerator is n and the denominator is the odd number $2n-1$.
2. Therefore $a_n = n/(2n-1)$.

Final answer

$$a_n = \frac{n}{2n-1}$$

Q068 Written

For the sequence $\{a_n\} : 1, \frac{3}{4}, \frac{5}{9}, \frac{7}{16}, \frac{9}{25}, \dots$, deduce the general term.

Solution

1. The numerator is the odd number $2n-1$ and the denominator is n^2 .
2. Therefore $a_n = (2n-1)/n^2$.

Final answer

$$a_n = \frac{2n-1}{n^2}$$

Worked Solutions

Arithmetic Sequences

Q069 Written

For the sequence $\{a_n\} : 1, -\frac{1}{3}, \frac{1}{5}, -\frac{1}{7}, \frac{1}{9}, \dots$, deduce the general term.

Solution

1. The denominator is $2n - 1$ and the signs alternate from a positive first term, so $a_n = (-1)^{n-1}/(2n - 1)$.

Final answer

$$a_n = \frac{(-1)^{n-1}}{2n - 1}$$

Q070 Written

For the sequence $\{a_n\} : \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots$, deduce the general term.

Solution

1. The numerator is n and the denominator is $n+1$, so $a_n = n/(n+1)$.

Final answer

$$a_n = \frac{n}{n + 1}$$

Worked Solutions

Arithmetic Sequences

Q071 Written

For the sequence $\{a_n\} : 1 \times 3, 2 \times 9, 3 \times 27, 4 \times 81, 5 \times 243, \dots$, deduce the general term.

Solution

1. The first factor is n and the second factor is 3^n , so $a_n = n3^n$.

Final answer

$$a_n = n 3^n$$

Q072 Written

For the sequence $\{a_n\} : 1 \times \frac{1}{2}, 2 \times \frac{1}{3}, 3 \times \frac{1}{4}, 4 \times \frac{1}{5}, 5 \times \frac{1}{6}, \dots$, deduce the general term then find the 20th term.

Solution

1. The first factor is the term number n and the second factor is $1/(n+1)$.
2. Hence the general term is $a_n = n/(n+1)$.
3. Substitute $n = 20$: $a_{20} = 20/(20+1) = 20/21$.

Final answer

$$a_n = \frac{n}{n+1}; \quad a_{20} = \frac{20}{21}$$

Worked Solutions

Arithmetic Sequences

Q073 Written

Find the first term and the common difference of the arithmetic sequences: (a) 2, 5, 8, 11, ...; (b) 3, -2, -7, -12, -17,

Solution

1. For (a), $5 - 2 = 8 - 5 = 11 - 8 = 3$, so $a = 2, d = 3$.
2. For (b), each successive difference is -5 , so $a = 3, d = -5$.

Final answer

(a) $a = 2, d = 3$ (b) $a = 3, d = -5$

Q074 Written

Find the first term and common difference of: (a) 1, 5, 9, 13, 17, ...; (b) 23, 19, 15, 11, 7,

Solution

1. (a) The constant increase is (4), so $a = 1, d = 4$.
2. (b) The constant change is -4 , so $a = 23, d = -4$.

Final answer

(a) $a = 1, d = 4$ (b) $a = 23, d = -4$

Worked Solutions

Arithmetic Sequences

Q075 MCQ

The listed terms $\{a_n\} = (4, 9, 14, 19, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = 4$.
2. Subtract consecutive terms: $d = 5$.

Correct option: B

Final answer

$$a = 4, d = 5$$

Q076 MCQ

The listed terms $\{a_n\} = (-7, -3, 1, 5, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = -7$.
2. Subtract consecutive terms: $d = 4$.

Correct option: A

Final answer

$$a = -7, d = 4$$

Worked Solutions

Arithmetic Sequences

Q077 MCQ

The listed terms $\{a_n\} = (18, 13, 8, 3, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = 18$.
2. Subtract consecutive terms: $d = -5$.

Correct option: C**Final answer**

$$a = 18, d = -5$$

Q078 MCQ

The listed terms $\{a_n\} = (12, 8, 4, 0, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = 12$.
2. Subtract consecutive terms: $d = -4$.

Correct option: D**Final answer**

$$a = 12, d = -4$$

Worked Solutions

Arithmetic Sequences

Q079 Written

Find the first term and common difference of: (a) 3, 5, 7, 9, 11, ...; (b) $-13, -9, -5, -1, 3, \dots$; (c) $-2, 1, 4, 7, 10, \dots$; (d) $10, 7, 4, 1, -2, \dots$; (e) $6, \frac{16}{3}, \frac{14}{3}, 4, \frac{10}{3}, \dots$; (f) $-4, -\frac{9}{2}, -5, -\frac{11}{2}, -6, \dots$.

Solution

1. Subtract consecutive terms in each list. The six constant differences are 2, 4, 3, $-3, -\frac{2}{3}, -\frac{1}{2}$, respectively; the first terms are the first entries.

Final answer

(a)(3, 2); (b)($-13, 4$); (c)($-2, 3$); (d)($10, -3$); (e)($6, -\frac{2}{3}$); (f)($-4, -\frac{1}{2}$).

Worked Solutions

Arithmetic Sequences

Q080 MCQ

The listed terms $\{a_n\} = (\frac{1}{2}, \frac{5}{6}, \frac{7}{6}, \frac{3}{2}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = \frac{1}{2}$.
2. Subtract consecutive terms: $d = \frac{1}{3}$.

Correct option: B**Final answer**

$$a = \frac{1}{2}, d = \frac{1}{3}$$

Q081 MCQ

The listed terms $\{a_n\} = (-\frac{3}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{4}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = -\frac{3}{4}$.
2. Subtract consecutive terms: $d = \frac{1}{2}$.

Correct option: A**Final answer**

$$a = -\frac{3}{4}, d = \frac{1}{2}$$

Worked Solutions

Arithmetic Sequences

Q082 MCQ

The listed terms $\{a_n\} = (\frac{5}{3}, 2, \frac{7}{3}, \frac{8}{3}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = \frac{5}{3}$.
2. Subtract consecutive terms: $d = \frac{1}{3}$.

Correct option: C

Final answer

$$a = \frac{5}{3}, d = \frac{1}{3}$$

Q083 MCQ

The listed terms $\{a_n\} = (-\frac{5}{2}, -1, \frac{1}{2}, 2, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = -\frac{5}{2}$.
2. Subtract consecutive terms: $d = \frac{3}{2}$.

Correct option: B

Final answer

$$a = -\frac{5}{2}, d = \frac{3}{2}$$

Worked Solutions

Arithmetic Sequences

Q084 Written

The sequences are arithmetic. Find the common difference and the numbers that fit in (A) and (B): (a) $A, 2, -1, B, \dots$; (b) $13, A, B, -2, \dots$

Solution

- (a) $d = -1 - 2 = -3$; hence $A = 2 - d = 5$ and $B = -1 + d = -4$.
- (b) The change from 13 to -2 covers three equal steps: $3d = -15$, so $d = -5$; therefore $A = 8, B = 3$.

Final answer

$$(a) \quad d = -3, A = 5, B = -4 \quad (b) \quad d = -5, A = 8, B = 3$$

Q085 Written

The sequences are arithmetic. Find the common difference and the missing numbers: (a) $2, 9, A, B, \dots$; (b) $A, 17, B, 5, \dots$

Solution

- (a) $d = 9 - 2 = 7$, so $A = 16, B = 23$.
- (b) $5 - 17 = 2d$, hence $d = -6$; then $A = 23, B = 11$.

Final answer

$$(a) \quad d = 7, A = 16, B = 23 \quad (b) \quad d = -6, A = 23, B = 11$$

Worked Solutions

Arithmetic Sequences

Q086 MCQ

The sequence $\{a_n\} = (6, A, 14, B, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Solution

1. Set the equal-step difference (d).
2. Substitute successively to obtain $d = 4, A = 10, B = 18$.

Correct option: A

Final answer

$$d = 4, A = 10, B = 18$$

Q087 MCQ

The sequence $\{a_n\} = (A, 5, 9, B, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Solution

1. Set the equal-step difference (d).
2. Substitute successively to obtain $d = 4, A = 1, B = 13$.

Correct option: C

Final answer

$$d = 4, A = 1, B = 13$$

Worked Solutions

Arithmetic Sequences

Q088 MCQ

The sequence $\{a_n\} = (11, A, B, 23, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Solution

1. Set the equal-step difference (d).
2. Substitute successively to obtain $d = 4, A = 15, B = 19$.

Correct option: B**Final answer**

$$d = 4, A = 15, B = 19$$

Worked Solutions

Arithmetic Sequences

Q089 Written

Find the common difference and missing terms: (a) $8, 15, A, B, \dots$; (b) $7, 13, A, B, \dots$; (c) $A, 9, 4, B, \dots$; (d) $A, 15, B, 9, \dots$; (e) $11, A, B, 20, \dots$; (f) $17, A, B, 5, C, \dots$

Solution

1. Use the equal-step condition. For example, in (d), $9 - 15 = 2d$, so $d = -2$, giving $A = 17, B = 13$. Applying the same calculation to (a)–(f) gives the listed values.

Final answer

(a) $d = 7, A = 22, B = 29$; (b) $d = 6, A = 19, B = 25$; (c) $d = -5, A = 14, B = -1$; (d) $d = -2, A = 17, B = 13$; (e) $d = 3, A = 14, B = 17$; (f) $d = -4, A = 13, B = 9, C = 1$.

Worked Solutions

Arithmetic Sequences

Q090 Written

For each arithmetic sequence, find the general term and the 7th term: (a) first term (4), common difference (3); (b) first term (2), common difference -3 .

Solution

1. (a) $a_n = 4 + 3(n - 1) = 3n + 1$; $a_7 = 22$.
2. (b) $a_n = 2 - 3(n - 1) = 5 - 3n$; $a_7 = -16$.

Final answer

$$(a) \quad a_n = 3n + 1, a_7 = 22 \quad (b) \quad a_n = 5 - 3n, a_7 = -16$$

Q091 Written

For each arithmetic sequence, find the general term and the 7th term: (a) $a = 5, d = 4$; (b) $a = -2, d = 3$; (c) $a = 10, d = -3$; (d) $a = -\frac{1}{2}, d = -\frac{1}{2}$.

Solution

1. Substitute each pair into $a_n = a + (n - 1)d$, then put $n = 7$. The results are $4n + 1, 3n - 5, 13 - 3n, -\frac{n}{2}$, with 7th terms 29, 16, $-8, -\frac{7}{2}$.

Final answer

$$(a) a_n = 4n + 1, a_7 = 29; (b) a_n = 3n - 5, a_7 = 16; (c) a_n = 13 - 3n, a_7 = -8; (d) a_n = -\frac{n}{2}, a_7 = -\frac{7}{2}.$$

Worked Solutions

Arithmetic Sequences

Q092 Written

For the arithmetic sequence a_n with first term (8) and common difference -6 , find the general term and the 20th term.

Solution

1. Use $a_n = a + (n - 1)d$: $a_n = 8 - 6(n - 1) = 14 - 6n$.
2. Substitute $n = 20$: $a_{20} = 14 - 120 = -106$.

Final answer

$$a_n = 8 + (n - 1)(-6) = 14 - 6n; \quad a_{20} = -106$$

Q093 Written

For each arithmetic sequence, find the general term and the 10th term: (a) $a = -7, d = 3$; (b) $a = -2, d = -3$; (c) $a = 7, d = \frac{1}{2}$; (d) $a = \frac{2}{3}, d = -\frac{1}{3}$.

Solution

1. Use $a_n = a + (n - 1)d$ and substitute $n = 10$. Simplification gives the four displayed formulas and terms.

Final answer

$$(a)a_n = 3n - 10, a_{10} = 20; (b)a_n = 1 - 3n, a_{10} = -29; (c)a_n = \frac{n + 13}{2}, a_{10} = \frac{23}{2}; (d)a_n = \frac{3 - n}{3}, a_{10} = -\frac{7}{3}.$$

Worked Solutions

Arithmetic Sequences

Q094 Written

In an arithmetic sequence, $a_3 + a_7 = 38$ and $\frac{a_4}{a_{12}} = \frac{2}{5}$. Find the first term and common difference.

Solution

1. $a_3 + a_7 = (a + 2d) + (a + 6d) = 2a + 8d = 38$, so $a + 4d = 19$.
2. The ratio gives $5(a + 3d) = 2(a + 11d)$, hence $3a = 7d$.
3. Solving yields $d = 3$ and $a = 7$.

Final answer

$$a = 7, d = 3.$$

Q095 Written

Find the general term of the arithmetic sequence whose first term is -5 and whose 17th term is 75.

Solution

1. $75 = -5 + 16d$, so $d = 5$.
2. $a_n = -5 + 5(n - 1) = 5n - 10$.

Final answer

$$a_n = 5n - 10$$

Worked Solutions

Arithmetic Sequences

Q096 Written

The first term is 6, and the 4th term is 18. Find a_n .

Solution

1. $18 = 6 + 3d$, so $d = 4$.
2. $a_n = 6 + 4(n - 1) = 4n + 2$.

Final answer

$$a_n = 4n + 2$$

Q097 Written

The first term is 20, and the 8th term is -1 . Find a_n .

Solution

1. $-1 = 20 + 7d$, so $d = -3$.
2. $a_n = 20 - 3(n - 1) = 23 - 3n$.

Final answer

$$a_n = 23 - 3n$$

Worked Solutions

Arithmetic Sequences

Q098 Written

The first term is -8 , and the 8th term is 55 . Find a_n .

Solution

1. $55 = -8 + 7d$, so $d = 9$.
2. $a_n = -8 + 9(n - 1) = 9n - 17$.

Final answer

$$a_n = 9n - 17$$

Q099 Written

The first term is -18 , and the 10th term is -63 . Find a_n .

Solution

1. $-63 = -18 + 9d$, so $d = -5$.
2. $a_n = -18 - 5(n - 1) = -5n - 13$.

Final answer

$$a_n = -5n - 13$$

Worked Solutions

Arithmetic Sequences

Q100 MCQ

An arithmetic sequence has $a = 4$ and $a_7 = 22$. Select the correct pair (a_n, d) :

Solution

1. $22 = 4 + 6d$, so $d = 3$.
2. $a_n = 4 + 3(n - 1) = 3n + 1$.

Correct option: D

Final answer

$$(a_n, d) = (3n + 1, 3)$$

Q101 MCQ

An arithmetic sequence has $a = -3$ and $a_9 = 29$. Select the correct pair (a_n, d) :

Solution

1. $29 = -3 + 8d$, so $d = 4$.
2. $a_n = -3 + 4(n - 1) = 4n - 7$.

Correct option: B

Final answer

$$(a_n, d) = (4n - 7, 4)$$

Worked Solutions

Arithmetic Sequences

Q102 MCQ

An arithmetic sequence has $a = 12$ and $a_5 = -4$. Select the correct pair (a_n, d) :

Solution

1. $-4 = 12 + 4d$, so $d = -4$.
2. $a_n = 12 - 4(n - 1) = 16 - 4n$.

Correct option: A

Final answer

$$(a_n, d) = (16 - 4n, -4)$$

Q103 MCQ

An arithmetic sequence has $a = 5$ and $a_{11} = 45$. Select the correct pair (a_n, d) :

Solution

1. $45 = 5 + 10d$, so $d = 4$.
2. $a_n = 5 + 4(n - 1) = 4n + 1$.

Correct option: B

Final answer

$$(a_n, d) = (4n + 1, 4)$$

Worked Solutions

Arithmetic Sequences

Q104 Written

Find the general term of the arithmetic sequence whose common difference is 3 and whose 16th term is 81.

Solution

1. $81 = a + 15(3)$, so $a = 36$.
2. $a_n = 36 + 3(n - 1) = 3n + 33$.

Final answer

$$a_n = 3n + 33$$

Q105 Written

The common difference is 2, and the 8th term is 4. Find a_n .

Solution

1. $4 = a + 7(2)$, so $a = -10$.
2. $a_n = -10 + 2(n - 1) = 2n - 12$.

Final answer

$$a_n = 2n - 12$$

Worked Solutions

Arithmetic Sequences

Q106 Written

The common difference is -4 , and the 5th term is -7 . Find a_n .

Solution

1. $-7 = a + 4(-4)$, so $a = 9$.
2. $a_n = 9 - 4(n - 1) = 13 - 4n$.

Final answer

$$a_n = 13 - 4n$$

Q107 Written

The common difference is -7 , and the 6th term is -24 . Find a_n .

Solution

1. $-24 = a + 5(-7)$, so $a = 11$.
2. $a_n = 11 - 7(n - 1) = 18 - 7n$.

Final answer

$$a_n = 18 - 7n$$

Worked Solutions

Arithmetic Sequences

Q108 MCQ

An arithmetic sequence has common difference -2 and $a_4 = 1$. Select the correct pair (a_n, a) :

Solution

1. $1 = a + 3(-2)$, so $a = 7$.
2. $a_n = 7 - 2(n - 1) = 9 - 2n$.

Correct option: A

Final answer

$$(a_n, a) = (9 - 2n, 7)$$

Q109 MCQ

An arithmetic sequence has common difference 5 and $a_6 = 17$. Select the correct pair (a_n, a) :

Solution

1. $17 = a + 5(5)$, so $a = -8$.
2. $a_n = -8 + 5(n - 1) = 5n - 13$.

Correct option: D

Final answer

$$(a_n, a) = (5n - 13, -8)$$

Worked Solutions

Arithmetic Sequences

Q110 MCQ

An arithmetic sequence has common difference 4 and $a_{10} = 38$. Select the correct pair (a_n, a) :

Solution

1. $38 = a + 9(4)$, so $a = 2$.
2. $a_n = 2 + 4(n - 1) = 4n - 2$.

Correct option: C

Final answer

$$(a_n, a) = (4n - 2, 2)$$

Q111 MCQ

An arithmetic sequence has common difference -3 and $a_8 = -18$. Select the correct pair (a_n, a) :

Solution

1. $-18 = a + 7(-3)$, so $a = 3$.
2. $a_n = 3 - 3(n - 1) = 6 - 3n$.

Correct option: D

Final answer

$$(a_n, a) = (6 - 3n, 3)$$

Worked Solutions

Arithmetic Sequences

Q112 Written

Find the general term of the arithmetic sequence $\{a_n\}$ whose 3rd term is 10 and the 8th term is 50.

Solution

1. Use $a_n = a + (n - 1)d$. The data give $10 = a + 2d$ and $50 = a + 7d$.
2. Subtracting gives $5d = 40$, so $d = 8$ and $a = -6$.
3. Therefore $a_n = -6 + 8(n - 1) = 8n - 14$.

Final answer

$$a_n = 8n - 14$$

Q113 Written

Find the general term of the arithmetic sequence whose 6th term is 2 and whose 15th term is -25 .

Solution

1. $2 = a + 5d$ and $-25 = a + 14d$, so $d = -3$, $a = 17$.
2. $a_n = 17 - 3(n - 1) = 20 - 3n$.

Final answer

$$a_n = 20 - 3n$$

Worked Solutions

Arithmetic Sequences

Q114 **Written****The 4th term is 14, and the 10th term is 62. Find a_n .****Solution**

1. $14 = a + 3d$ and $62 = a + 9d$, so $d = 8$, $a = -10$.
2. $a_n = 8n - 18$.

Final answer

$$a_n = 8n - 18$$

Q115 **Written****The 3rd term is 30, and the 10th term is 2. Find a_n .****Solution**

1. $30 = a + 2d$ and $2 = a + 9d$, so $d = -4$, $a = 38$.
2. $a_n = 42 - 4n$.

Final answer

$$a_n = 42 - 4n$$

Worked Solutions

Arithmetic Sequences

Q116 Written

The 10th term is 49, and the 23rd term is 10. Find a_n .

Solution

1. $49 = a + 9d$ and $10 = a + 22d$, so $d = -3$, $a = 76$.
2. $a_n = 79 - 3n$.

Final answer

$$a_n = 79 - 3n$$

Q117 Written

The 3rd term is 6, and the 10th term is 62. Find a_n .

Solution

1. $6 = a + 2d$ and $62 = a + 9d$, so $d = 8$, $a = -10$.
2. $a_n = 8n - 18$.

Final answer

$$a_n = 8n - 18$$

Worked Solutions

Arithmetic Sequences

Q118 MCQ

An arithmetic sequence has $a_3 = 5$ and $a_9 = 17$. Select the correct pair (a_n, a) :

Solution

1. $5 = a + 2d$ and $17 = a + 8d$, so $d = 2$, $a = 1$.
2. $a_n = 1 + 2(n - 1) = 2n - 1$.

Correct option: C

Final answer

$$(a_n, a) = (2n - 1, 1)$$

Q119 MCQ

An arithmetic sequence has $a_4 = 7$ and $a_{10} = -5$. Select the correct pair (a_n, a) :

Solution

1. $7 = a + 3d$ and $-5 = a + 9d$, so $d = -2$, $a = 13$.
2. $a_n = 13 - 2(n - 1) = 15 - 2n$.

Correct option: C

Final answer

$$(a_n, a) = (15 - 2n, 13)$$

Worked Solutions

Arithmetic Sequences

Q120 MCQ

An arithmetic sequence has $a_7 = 31$ and $a_{12} = 56$. Select the correct pair (a_n, a) :

Solution

1. $31 = a + 6d$ and $56 = a + 11d$, so $d = 5$, $a = 1$.

2. $a_n = 1 + 5(n - 1) = 5n - 4$.

Correct option: B

Final answer

$$(a_n, a) = (5n - 4, 1)$$

Q121 MCQ

An arithmetic sequence has $a_2 = -4$ and $a_8 = 14$. Select the correct pair (a_n, a) :

Solution

1. $-4 = a + d$ and $14 = a + 7d$, so $d = 3$, $a = -7$.

2. $a_n = -7 + 3(n - 1) = 3n - 10$.

Correct option: A

Final answer

$$(a_n, a) = (3n - 10, -7)$$

Worked Solutions

Arithmetic Sequences

Q122 MCQ

An arithmetic sequence has $a_5 = 3$ and $a_{11} = 9$. Select the correct pair (a_n, a) :

Solution

1. $3 = a + 4d$ and $9 = a + 10d$, so $d = \frac{1}{2}$, $a = -\frac{1}{2}$.

2. $a_n = -\frac{1}{2} + \frac{1}{2}(n-1) = \frac{n-2}{2}$.

Correct option: A

Final answer

$$\left(a_n = \frac{n-2}{2}, -\frac{1}{2}\right)$$

Q123 MCQ

An arithmetic sequence has $a_2 = 6$ and $a_9 = 20$. Select the correct pair (a_n, a) :

Solution

1. $6 = a + d$ and $20 = a + 8d$, so $d = 2$, $a = 4$.

2. $a_n = 4 + 2(n-1) = 2n + 2$.

Correct option: B

Final answer

$$(a_n = 2n + 2, 4)$$

Worked Solutions

Arithmetic Sequences

Q124 Written

The common difference is $\frac{2}{5}$, and the 30th term is $\frac{23}{5}$. Find a_n .

Solution

$$1. \frac{23}{5} = a + 29\left(\frac{2}{5}\right), \text{ so } a = -7.$$

$$2. a_n = -7 + \frac{2}{5}(n - 1) = \frac{2n - 37}{5}.$$

Final answer

$$a_n = \frac{2n - 37}{5}$$

Q125 MCQ

An arithmetic sequence has $d = -\frac{3}{2}$ and $a_5 = -9$. Select the correct pair (a_n, a) :

Solution

$$1. -9 = a + 4\left(-\frac{3}{2}\right), \text{ so } a = -3.$$

$$2. a_n = -3 - \frac{3}{2}(n - 1) = -\frac{3(n+1)}{2}.$$

Correct option: A

Final answer

$$\left(a_n = -\frac{3(n+1)}{2}, -3\right)$$

Worked Solutions

Arithmetic Sequences

Q126 MCQ

An arithmetic sequence has $d = \frac{2}{3}$ and $a_{12} = 10$. Select the correct pair (a_n, a) :

Solution

1. $10 = a + 11(\frac{2}{3})$, so $a = \frac{8}{3}$.
2. $a_n = \frac{8}{3} + \frac{2}{3}(n - 1) = \frac{2n+6}{3}$.

Correct option: D

Final answer

$$\left(a_n = \frac{2n+6}{3}, \frac{8}{3}\right)$$

Q127 Written

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, which term is 113?

Solution

1. First find a : $33 = a + 7(4)$, so $a = 5$.
2. Then $a_n = 5 + 4(n - 1) = 4n + 1$.
3. Solve $4n + 1 = 113$: $n = 28$.

Final answer

$n = 28$, the 28th term

Worked Solutions

Arithmetic Sequences

Q128 Written

For the arithmetic sequence with first term 1 and common difference 3, determine which term is 40.

Solution

1. $a_n = 1 + 3(n - 1) = 3n - 2$.
2. $3n - 2 = 40$ gives $n = 14$.

Final answer

$n = 14$, the 14th term

Q129 Written

For the arithmetic sequence with common difference -5 and 13th term -7 , determine which term is -52 .

Solution

1. $-7 = a + 12(-5)$, so $a = 53$ and $a_n = 58 - 5n$.
2. $58 - 5n = -52$ gives $n = 22$.

Final answer

$n = 22$, the 22nd term

Worked Solutions

Arithmetic Sequences

Q130 Written

For the arithmetic sequence whose 10th term is -20 and 20th term is -50 , determine which term is -110 .

Solution

1. $-20 = a + 9d$ and $-50 = a + 19d$, so $d = -3$, $a = 7$.
2. $a_n = 10 - 3n$, and $10 - 3n = -110$ gives $n = 40$.

Final answer

$n = 40$, the 40th term

Q131 MCQ

For $a = 3$, $d = 4$, which term of the arithmetic sequence is 59?:

Solution

1. $a_n = 3 + 4(n - 1) = 4n - 1$.
2. $4n - 1 = 59$ gives $n = 15$.

Correct option: C

Final answer

$n = 15$

Worked Solutions

Arithmetic Sequences

Q132 MCQ

For $a = -5, d = 6$, which term of the arithmetic sequence is 55?:

Solution

1. $a_n = -5 + 6(n - 1) = 6n - 11$.
2. $6n - 11 = 55$ gives $n = 11$.

Correct option: D

Final answer

$$n = 11$$

Q133 MCQ

For $a = 20, d = -3$, which term of the arithmetic sequence is -16 ?:

Solution

1. $a_n = 20 - 3(n - 1) = 23 - 3n$.
2. $23 - 3n = -16$ gives $n = 13$.

Correct option: A

Final answer

$$n = 13$$

Worked Solutions

Arithmetic Sequences

Q134 MCQ

For $a = 1$, $d = 5$, which term of the arithmetic sequence is 86?:

Solution

1. $a_n = 1 + 5(n - 1) = 5n - 4$.
2. $5n - 4 = 86$ gives $n = 18$.

Correct option: B

Final answer

$$n = 18$$

Q135 Written

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, find the first term greater than 500.

Solution

1. From the given data, $a_n = 4n + 1$.
2. Require $4n + 1 > 500$, hence $n > 124.75$.
3. The least natural index is $n = 125$.

Final answer

$n = 125$, the 125th term

Worked Solutions

Arithmetic Sequences

Q136 Written

For the arithmetic sequence with first term 1 and common difference 3, find the first term greater than 100.

Solution

1. $a_n = 3n - 2$.
2. $3n - 2 > 100$ gives $n > 34$.
3. The first valid index is $n = 35$.

Final answer

$n = 35$, the 35th term

Q137 MCQ

For $a = 2, d = 7$, find the first term greater than 100:

Solution

1. $a_n = 7n - 5$.
2. $7n - 5 > 100$ gives $n > 15$, so $n = 16$.

Correct option: A

Final answer

$n = 16$

Worked Solutions

Arithmetic Sequences

Q138 MCQ

For $a = -9, d = 4$, find the first term greater than 50:**Solution**

1. $a_n = 4n - 13$.
2. $4n - 13 > 50$ gives $n > 15.75$, so $n = 16$.

Correct option: D**Final answer**

$$n = 16$$

Q139 MCQ

For $a = 6, d = 3$, find the first term greater than 200:**Solution**

1. $a_n = 3n + 3$.
2. $3n + 3 > 200$ gives $n > 65\frac{2}{3}$, so $n = 66$.

Correct option: D**Final answer**

$$n = 66$$

Worked Solutions

Arithmetic Sequences

Q140 MCQ

For $a = -1$, $d = 9$, find the first term greater than 300:

Solution

1. $a_n = 9n - 10$.
2. $9n - 10 > 300$ gives $n > 34\frac{4}{9}$, so $n = 35$.

Correct option: B

Final answer

$$n = 35$$

Q141 Written

For the arithmetic sequence with common difference -5 and 13th term -7 , find the first term smaller than -300 .

Solution

1. $a_n = 58 - 5n$.
2. $58 - 5n < -300$ gives $n > 71.6$.
3. The first valid index is $n = 72$.

Final answer

$n = 72$, the 72nd term

Worked Solutions

Arithmetic Sequences

Q142 Written

For the arithmetic sequence whose 10th term is -20 and 20th term is -50 , find the first term smaller than -500 .

Solution

1. $a_n = 10 - 3n$.
2. $10 - 3n < -500$ gives $n > 170$.
3. The first valid index is $n = 171$.

Final answer

$n = 171$, the 171st term

Q143 MCQ

For $a = 10$, $d = -4$, find the first term smaller than -100 :

Solution

1. $a_n = 14 - 4n$.
2. $14 - 4n < -100$ gives $n > 28.5$, so $n = 29$.

Correct option: B

Final answer

$n = 29$

Worked Solutions

Arithmetic Sequences

Q144 MCQ

For $a = 25, d = -6$, find the first term smaller than -250 :**Solution**

1. $a_n = 25 - 6n$.
2. $25 - 6n < -250$ gives $n > 46\frac{5}{6}$, so $n = 47$.

Correct option: B**Final answer**

$$n = 47$$

Q145 MCQ

For $a = -3, d = -5$, find the first term smaller than -200 :**Solution**

1. $a_n = -3 - 5n$.
2. $-3 - 5n < -200$ gives $n > 40.4$, so $n = 41$.

Correct option: C**Final answer**

$$n = 41$$

Worked Solutions

Arithmetic Sequences

Q146 MCQ

For $a = 4$, $d = 6$, find the first term that is at least 100:

Solution

1. $a_n = 6n - 2$.
2. $6n - 2 \geq 100$ gives $n \geq 17$, so $n = 17$.

Correct option: A

Final answer

$$n = 17$$

Q147 MCQ

For the expression For $a = 5$, $d = 4$, does the sequence contain the term 42?, the required value is:

Solution

1. $a_n = 4n + 1$.
2. $4n + 1 = 42$ gives $n = \frac{41}{4}$, not a natural index.
3. Therefore 42 is not a term.

Correct option: B

Final answer

No term

Worked Solutions

Arithmetic Sequences

Q148 **Written**

When $a, 6, 2a$ form an arithmetic sequence in this order, find a .

Solution

1. $2(6) = a + 2a$.
2. $3a = 12$, so $a = 4$.

Final answer

$$a = 4$$

Q149 **Written**

When the three numbers $x, 6, x^2$ form an arithmetic sequence in this order, find x .

Solution

1. Use $2b = a + c$: $2(6) = x + x^2$.
2. Thus $x^2 + x - 12 = 0 = (x + 4)(x - 3)$.
3. Therefore $x = -4$ or $x = 3$.

Final answer

$$x = -4 \text{ or } x = 3$$

Worked Solutions

Arithmetic Sequences

Q150 **Written****When $x^2, x, -8$ form an arithmetic sequence in this order, find x .****Solution**

1. $2x = x^2 - 8$.
2. $x^2 - 2x - 8 = (x - 4)(x + 2) = 0$.
3. $x = -2$ or $x = 4$.

Final answer

$$x = -2 \text{ or } x = 4$$

Q151 **Written****When $-3, x, x^2$ form an arithmetic sequence in this order, find x .****Solution**

1. $2x = -3 + x^2$.
2. $x^2 - 2x - 3 = (x - 3)(x + 1) = 0$.
3. $x = -1$ or $x = 3$.

Final answer

$$x = -1 \text{ or } x = 3$$

Worked Solutions

Arithmetic Sequences

Q152 Challenge

When $2, x, y, z$ form an arithmetic sequence in this order, and $2x, y + 1, z - y$ also form an arithmetic sequence in this order, find x, y, z .

Solution

1. Let the common difference of $2, x, y, z$ be d : $x = 2 + d$, $y = 2 + 2d$, $z = 2 + 3d$.
2. $2(y + 1) = 2x + (z - y)$ gives $6 + 4d = 4 + 3d$, hence $d = -2$.
3. $(x, y, z) = (0, -2, -4)$.

Final answer

$$x = 0, \quad y = -2, \quad z = -4$$

Worked Solutions

Arithmetic Sequences

Q153 Written

Find the sum (S_7) of a sequence with first term (8), last term (22), and (7) terms.

Solution

1. Use ($S_n = \frac{n}{2}(a + l)$).
2. ($S_7 = \frac{7}{2}(8 + 22) = 105$).

Final answer

105

Q154 Written

Find (S_{10}) for a sequence with first term (7), last term (13), and (10) terms.

Solution

1. ($S_{10} = \frac{10}{2}(7 + 13) = 100$).

Final answer

100

Worked Solutions

Arithmetic Sequences

Q155 **Written****Find (S_{10}) for a sequence with first term (3), last term (21), and (10) terms.****Solution**

1. $(S_{10} = \frac{10}{2}(3 + 21) = 120).$

Final answer

120

Q156 **Written****Find (S_8) for a sequence with first term (5), last term (33), and (8) terms.****Solution**

1. $(S_8 = \frac{8}{2}(5 + 33) = 152).$

Final answer

152

Worked Solutions

Arithmetic Sequences

Q157 **Written****Find (S_{10}) for a sequence with first term (2), last term (29), and (10) terms.****Solution**

1. $(S_{10} = \frac{10}{2}(2 + 29) = 155).$

Final answer

155

Q158 **MCQ****A sequence has first term 6, last term 30, and 7 terms. Find its sum:****Solution**

1. Use $S_7 = \frac{7}{2}(6 + 30).$
2. Thus $S_7 = 126.$

Correct option: B**Final answer**

126

Worked Solutions

Arithmetic Sequences

Q159 MCQ

A sequence has first term -4 , last term 20 , and 9 terms. Find its sum:

Solution

1. $S_9 = \frac{9}{2}(-4 + 20) = 72.$

Correct option: B

Final answer

72

Q160 MCQ

Find the sum of 10 terms whose first term is 15 and last term is 3 :

Solution

1. $S_{10} = \frac{10}{2}(15 + 3) = 90.$

Correct option: B

Final answer

90

Worked Solutions

Arithmetic Sequences

Q161 MCQ

The first term is 7, the last term is 42, and there are 8 terms. Find S_8 :

Solution

1. $S_8 = \frac{8}{2}(7 + 42) = 196.$

Correct option: B

Final answer

196

Q162 MCQ

Find the sum of 6 terms with first term -12 and last term 8:

Solution

1. $S_6 = \frac{6}{2}(-12 + 8) = -12.$

Correct option: B

Final answer

-12

Worked Solutions

Arithmetic Sequences

Q163 MCQ

A sequence has $S_9 = 72$ and first term -4 . Find its last term:

Solution

1. From $72 = \frac{9}{2}(-4 + l)$, $l = 20$.

Correct option: C

Final answer

20

Q164 Written

Find (S_{13}) when the first term is (20) , the common difference is (-5) , and the number of terms is (13) .

Solution

1. Use $(S_n = \frac{n}{2}\{2a + (n - 1)d\})$.

2. $(S_{13} = \frac{13}{2}\{40 + 12(-5)\} = -130)$.

Final answer

-130

Worked Solutions

Arithmetic Sequences

Q165 **Written****Find (S_{12}) when the first term is (-6) , the common difference is (3) , and there are (12) terms.****Solution**

$$1. (S_{12} = \frac{12}{2}\{-12 + 11(3)\} = 144).$$

Final answer

144

Q166 **Written****Find (S_{15}) when the first term is (10) , the common difference is (-3) , and there are (15) terms.****Solution**

$$1. (S_{15} = \frac{15}{2}\{20 + 14(-3)\} = -165).$$

Final answer

-165

Worked Solutions

Arithmetic Sequences

Q167 Written

Find (S_{11}) when the first term is (-4) , the common difference is (4) , and there are (11) terms.

Solution

1. $(S_{11} = \frac{11}{2}\{-8 + 10(4)\} = 176).$

Final answer

176

Q168 Written

Find (S_{18}) when the first term is (3) , the common difference is (2) , and there are (18) terms.

Solution

1. $(S_{18} = \frac{18}{2}\{6 + 17(2)\} = 360).$

Final answer

360

Worked Solutions

Arithmetic Sequences

Q169 MCQ

For $a = 4, d = 3, n = 12$, find S_{12} :**Solution**

$$1. S_{12} = \frac{12}{2}\{8 + 11(3)\} = 246.$$

Correct option: C**Final answer**

246

Q170 MCQ

For $a = 9, d = -2, n = 10$, find S_{10} :**Solution**

$$1. S_{10} = 5\{18 + 9(-2)\} = 0.$$

Correct option: B**Final answer**

0

Worked Solutions

Arithmetic Sequences

Q171 MCQ

For $a = -5, d = 4, n = 9$, find S_9 :**Solution**

$$1. S_9 = \frac{9}{2}\{-10 + 8(4)\} = 88.$$

Correct option: C**Final answer**

88

Q172 MCQ

For $a = 2, d = -3, n = 15$, find S_{15} :**Solution**

$$1. S_{15} = \frac{15}{2}\{4 + 14(-3)\} = -285.$$

Correct option: B**Final answer**

-285

Worked Solutions

Arithmetic Sequences

Q173 MCQ

For $a = 11, d = 0, n = 8$, find S_8 :**Solution**1. All terms equal 11, so $S_8 = 88$.**Correct option:** C**Final answer**

88

Q174 MCQ

For $a = -3, d = -2, n = 7$, find S_7 :**Solution**1. $S_7 = \frac{7}{2}\{-6 + 6(-2)\} = -54$.**Correct option:** B**Final answer**

-54

Worked Solutions

Arithmetic Sequences

Q175 Written

Find the sum (S) of a sequence with first term (2), common difference (3), and last term (47).**Solution**

1. Find (n) from $(47 = 2 + (n - 1)3)$, giving ($n = 16$).
2. Then $(S = \frac{16}{2}(2 + 47) = 392)$.

Final answer

392

Q176 Written

Find (S) for a sequence with first term (5), common difference (3), and last term (44).**Solution**

1. Solve $(44 = 5 + (n - 1)3)$, so ($n = 14$).
2. $(S = \frac{14}{2}(5 + 44) = 343)$.

Final answer

343

Worked Solutions

Arithmetic Sequences

Q177 Written

Find (S) for a sequence with first term (3), common difference (4), and last term (75).**Solution**

1. $(75 = 3 + (n - 1)4)$ gives $(n = 19)$.
2. $(S = \frac{19}{2}(3 + 75) = 741)$.

Final answer

741

Q178 Written

Find (S) for a sequence with first term (2), common difference (5), and last term (57).**Solution**

1. $(57 = 2 + (n - 1)5)$ gives $(n = 12)$.
2. $(S = \frac{12}{2}(2 + 57) = 354)$.

Final answer

354

Worked Solutions

Arithmetic Sequences

Q179 Written

Find (S) for a sequence with first term (4), common difference (3), and last term (100).

Solution

1. $100 = 4 + (n - 1)3$ gives $(n = 33)$.
2. $S = \frac{33}{2}(4 + 100) = 1716$.

Final answer

1716

Q180 MCQ

A sequence has $a = 5, d = 4, l = 41$. Find its sum:

Solution

1. $41 = 5 + (n - 1)4$ gives $n = 10$.
2. $S = \frac{10}{2}(5 + 41) = 230$.

Correct option: C

Final answer

230

Worked Solutions

Arithmetic Sequences

Q181 MCQ

A sequence has $a = -7, d = 5, l = 48$. Find its sum:

Solution

1. $48 = -7 + (n - 1)5$ gives $n = 12$.
2. $S = 6(-7 + 48) = 246$.

Correct option: C

Final answer

246

Q182 MCQ

A sequence has $a = 12, d = 2, l = 50$. Find its sum:

Solution

1. $50 = 12 + (n - 1)2$ gives $n = 20$.
2. $S = 10(12 + 50) = 620$.

Correct option: C

Final answer

620

Worked Solutions

Arithmetic Sequences

Q183 MCQ

A sequence has $a = -10, d = 6, l = 50$. Find its sum:**Solution**

1. $50 = -10 + (n - 1)6$ gives $n = 11$.

2. $S = \frac{11}{2}(-10 + 50) = 220$.

Correct option: C**Final answer**

220

Q184 MCQ

A sequence has $a = 18, d = -3, l = -9$. Find its sum:**Solution**

1. Solve $-9 = 18 + (n - 1)(-3)$, giving $n = 10$.

2. $S = \frac{10}{2}(18 - 9) = 45$.

Correct option: C**Final answer**

45

Worked Solutions

Arithmetic Sequences

Q185 MCQ

A sequence has $a = 40, d = -4, l = -12$. Find its sum:

Solution

1. $n = 14$ from $-12 = 40 + (n - 1)(-4)$.
2. $S = 7(40 - 12) = 196$.

Correct option: B

Final answer

196

Q186 Written

For the arithmetic sequence with first term (48) and common difference (-2) , find the sum from the 20th term to the 30th term.

Solution

1. Use $(S_{30} - S_{19})$, since the range is (a_{20}) through (a_{30}) .
2. $(a_{20} = 10, a_{30} = -10)$, so $(11 \times \frac{10 + (-10)}{2} = 0)$.

Final answer

0

Worked Solutions

Arithmetic Sequences

Q187 Written

For the arithmetic sequence with first term (-53) and common difference (4) , find the sum from the 10th term to the 25th term.

Solution

1. $(a_{10} = -53 + 9(4) = -17)$ and $(a_{25} = -53 + 24(4) = 43)$.
2. There are (16) terms, so the sum is $(16 \times \frac{-17+43}{2} = 208)$.

Final answer

208

Q188 MCQ

For $a = 3, d = 4$, find the sum from a_5 to a_{12} :

Solution

1. $a_5 = 19, a_{12} = 47$, with 8 terms.
2. Sum = $8(19 + 47)/2 = 264$.

Correct option: B

Final answer

264

Worked Solutions

Arithmetic Sequences

Q189 MCQ

For $a = 20$, $d = -3$, find the sum from a_4 to a_{10} :**Solution**

1. $a_4 = 11$, $a_{10} = -7$, with 7 terms.
2. Sum = $7(11 - 7)/2 = 14$.

Correct option: C**Final answer**

14

Q190 MCQ

For $a = 60$, $d = -3$, find the sum from a_8 to a_{16} :**Solution**

1. $a_8 = 39$, $a_{16} = 15$, with 9 terms.
2. Sum = $9(39 + 15)/2 = 243$.

Correct option: C**Final answer**

243

Worked Solutions

Arithmetic Sequences

Q191 MCQ

For $a = 1, d = 7$, find the sum from a_3 to a_9 :**Solution**

1. $a_3 = 15, a_9 = 57$, with 7 terms.
2. $\text{Sum} = 7(15 + 57)/2 = 252$.

Correct option: C**Final answer**

252

Q192 MCQ

For $a = 60, d = -3$, find the sum from a_9 to a_{17} :**Solution**

1. $a_9 = 36, a_{17} = 12$, with 9 terms.
2. $\text{Sum} = 9(36 + 12)/2 = 216$.

Correct option: B**Final answer**

216

Worked Solutions

Arithmetic Sequences

Q193 MCQ

For $a = 12$, $d = 5$, find the sum from a_2 to a_8 :

Solution

1. $a_2 = 17$, $a_8 = 47$, with 7 terms.
2. Sum $= 7(17 + 47)/2 = 224$.

Correct option: C

Final answer

224

Q194 Written

For the arithmetic sequence $(90, 84, 78, \dots)$, find the sum from the 10th term to the 20th term.

Solution

1. Here $(a = 90, d = -6)$, so the required sum is $(S_{20} - S_9)$.
2. $(S_{20} = 660)$ and $(S_9 = 594)$, hence the sum is $(660 - 594 = 66)$.

Final answer

66

Worked Solutions

Arithmetic Sequences

Q195 Written

For the arithmetic sequence $(-9, -3, 3, \dots)$, find the sum from the 11th term to the 20th term.

Solution

1. Here $a = -9$, $d = 6$, so use $S_{20} - S_{10}$.
2. $a_{10} = 45$ and $a_{20} = 105$; there are 10 terms.
3. Required sum = $\frac{10(45+105)}{2} = 750$.

Final answer

750

Q196 Written

For the arithmetic sequence $(74, 69, 64, \dots)$, find the sum from the 8th term to the 18th term.

Solution

1. Here $(a = 74, d = -5)$, so $(a_8 = 39)$ and $(a_{18} = -11)$.
2. There are (11) terms; the sum is $(11 \times \frac{39-11}{2} = 154)$.

Final answer

154

Worked Solutions

Arithmetic Sequences

Q197 MCQ

For $a = 4, d = 5$, find the sum from a_6 to a_{13} :**Solution**

1. $a_6 = 29, a_{13} = 64$, with 8 terms.
2. $\text{Sum} = 8(29 + 64)/2 = 372$.

Correct option: C**Final answer**

372

Q198 MCQ

For $a = -25, d = 6$, find the sum from a_5 to a_{11} :**Solution**

1. $a_5 = -1, a_{11} = 35$, with 7 terms.
2. $\text{Sum} = 7(-1 + 35)/2 = 119$.

Correct option: C**Final answer**

119

Worked Solutions

Arithmetic Sequences

Q199 Written

Find (S_n) for a sequence with first term (1) and common difference (4).**Solution**

1. Substitute $(a = 1, d = 4)$ into $(S_n = \frac{n}{2}\{2a + (n - 1)d\})$.
2. $(S_n = \frac{n}{2}(2 + 4n - 4) = 2n^2 - n)$.

Final answer

$$2n^2 - n$$

Q200 Written

Find (S_n) for a sequence with first term (2) and common difference (-3) .**Solution**

1. $(S_n = \frac{n}{2}\{4 + (n - 1)(-3)\})$.
2. Therefore $(S_n = \frac{n(7-3n)}{2})$.

Final answer

$$\frac{n(7 - 3n)}{2}$$

Worked Solutions

Arithmetic Sequences

Q201 Written

Find (S_n) for a sequence with first term (0) and common difference (5).**Solution**

1. $(S_n = \frac{n}{2}\{0 + 5(n - 1)\})$.
2. Hence $(S_n = \frac{5n(n-1)}{2})$.

Final answer

$$\frac{5n(n-1)}{2}$$

Q202 Written

Find (S_n) for a sequence with first term (3) and common difference (-4) .**Solution**

1. $(S_n = \frac{n}{2}\{6 - 4(n - 1)\})$.
2. Therefore $(S_n = 5n - 2n^2)$.

Final answer

$$5n - 2n^2$$

Worked Solutions

Arithmetic Sequences

Q203 Written

Find (S_n) for a sequence with first term (-4) and common difference (-2) .**Solution**

1. $(S_n = \frac{n}{2}\{-8 - 2(n - 1)\})$.
2. Therefore $(S_n = -n^2 - 3n)$.

Final answer

$$-n^2 - 3n$$

Worked Solutions

Arithmetic Sequences

Q204 Challenge

For the arithmetic sequence $(94, 89, 84, \dots)$, find the sum of the odd-numbered terms from the first term to the 11th term.

Solution

1. The odd-numbered terms are $(a_1, a_3, a_5, a_7, a_9, a_{11})$, a new arithmetic sequence with first term (94), common difference $(2(-5) = -10)$, and (6) terms.
2. Its last term is (44), so the sum is $(\frac{6}{2}(94 + 44) = 414)$.

Final answer

414

Worked Solutions

Arithmetic Sequences

Q205 Written

Find the sum of the natural numbers from 1 to 100.**Solution**

1. There are $n = 100$ terms, with $a = 1$ and $l = 100$.
2. $S_{100} = \frac{100}{2}(1 + 100) = 5050$.

Final answer

5050

Q206 Written

Find the sum of the natural numbers from 1 to 200.**Solution**

1. Use $n = 200$, $a = 1$, $l = 200$.
2. $S = \frac{200}{2}(1 + 200) = 20100$.

Final answer

20100

Worked Solutions

Arithmetic Sequences

Q207 **Written****Find the sum of the natural numbers from 1 to 150.****Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 150$.
2. *Therefore* $S = 11325$.

Final answer

11325

Q208 **Written****Find the sum of the natural numbers from 1 to 70.****Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 70$.
2. *Therefore* $S = 2485$.

Final answer

2485

Worked Solutions

Arithmetic Sequences

Q209 Written

Find the sum of the natural numbers from 1 to 100.

Solution

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 100$.
2. Therefore $S = 5050$.

Final answer

5050

Q210 Written

Find the sum of the natural numbers from 1 to 400.

Solution

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 400$.
2. Therefore $S = 80200$.

Final answer

80200

Worked Solutions

Arithmetic Sequences

Q211 **Written****Find the sum of the natural numbers from 1 to 250.****Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 250$.
2. *Therefore* $S = 31375$.

Final answer

31375

Q212 **Written****Find the sum of the natural numbers from 1 to 81.****Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 81$.
2. *Therefore* $S = 8085$.

Final answer

8085

Worked Solutions

Arithmetic Sequences

Q213 **Written****Find the sum of the natural numbers from 1 to 301.****Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 301$.
2. *Therefore* $S = 24535$.

Final answer

24535

Q214 **Written****Find the sum of the natural numbers from 1 to 99.****Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 99$.
2. *Therefore* $S = 4950$.

Final answer

4950

Worked Solutions

Arithmetic Sequences

Q215 MCQ

Find the sum of the natural numbers from 1 to 30:**Solution**

$$1. S = \frac{30}{2}(1 + 30) = 465.$$

Correct option: B**Final answer**

465

Q216 MCQ

Find the sum of the natural numbers from 1 to 45:**Solution**

$$1. S = \frac{45}{2}(1 + 45) = 1035.$$

Correct option: B**Final answer**

1035

Worked Solutions

Arithmetic Sequences

Q217 MCQ

Find the sum of the natural numbers from 1 to 60:**Solution**

$$1. S = \frac{60}{2}(1 + 60) = 1830.$$

Correct option: B

Final answer

1830

Q218 MCQ

Find the sum of the natural numbers from 1 to 75:**Solution**

$$1. S = \frac{75}{2}(1 + 75) = 2850.$$

Correct option: B

Final answer

2850

Worked Solutions

Arithmetic Sequences

Q219 MCQ

Find the sum of the natural numbers from 1 to 90:**Solution**

$$1. S = \frac{90}{2}(1 + 90) = 4095.$$

Correct option: B**Final answer**

4095

Q220 MCQ

Find the sum of the natural numbers from 1 to 125:**Solution**

$$1. S = \frac{125}{2}(1 + 125) = 7875.$$

Correct option: B**Final answer**

7875

Worked Solutions

Arithmetic Sequences

Q221 MCQ

Find the sum of the natural numbers from 1 to 180:**Solution**

$$1. S = \frac{180}{2}(1 + 180) = 16290.$$

Correct option: B

Final answer

16290

Q222 MCQ

Find the sum of the natural numbers from 1 to 220:**Solution**

$$1. S = \frac{220}{2}(1 + 220) = 24310.$$

Correct option: B

Final answer

24310

Worked Solutions

Arithmetic Sequences

Q223 MCQ

Find the sum of the natural numbers from 1 to 275:**Solution**

$$1. S = \frac{275}{2}(1 + 275) = 37950.$$

Correct option: B**Final answer**

37950

Q224 MCQ

Find the sum of the natural numbers from 1 to 350:**Solution**

$$1. S = \frac{350}{2}(1 + 350) = 61425.$$

Correct option: B**Final answer**

61425

Worked Solutions

Arithmetic Sequences

Q225 Written

Find the sum of the natural numbers from 101 to 200.**Solution**

1. There are $n = 100$ terms, with $a = 101$ and $l = 200$.
2. $S = \frac{100}{2}(101 + 200) = 15050$.

Final answer

15050

Q226 Written

Find the sum of the natural numbers from 201 to 300.**Solution**

1. There are 100 terms from 201 to 300.
2. $S = \frac{100}{2}(201 + 300) = 25050$.

Final answer

25050

Worked Solutions

Arithmetic Sequences

Q227 **Written****Find the sum of all two-digit natural numbers.****Solution**

1. Use $10 + 11 + \cdots + 99 : n = 90$.
2. $S = \frac{90}{2}(10 + 99) = 4905$.

Final answer

4905

Q228 **MCQ****A sequence has first term 5, last term 35, and 7 terms. Select its sum:****Solution**

1. $S = \frac{7}{2}(5 + 35) = 140$.

Correct option: B

Final answer

140

Worked Solutions

Arithmetic Sequences

Q229 MCQ

A sequence has first term 12, last term 68, and 9 terms. Select its sum:

Solution

$$1. S = \frac{9}{2}(12 + 68) = 360.$$

Correct option: B

Final answer

360

Q230 MCQ

A sequence has first term -4, last term 20, and 9 terms. Select its sum:

Solution

$$1. S = \frac{9}{2}(-4 + 20) = 72.$$

Correct option: B

Final answer

72

Worked Solutions

Arithmetic Sequences

Q231 MCQ

A sequence has first term 7, last term 42, and 8 terms. Select its sum:

Solution

1. $S = \frac{8}{2}(7 + 42) = 196.$

Correct option: B

Final answer

196

Q232 MCQ

A sequence has first term 15, last term 3, and 10 terms. Select its sum:

Solution

1. $S = \frac{10}{2}(15 + 3) = 90.$

Correct option: B

Final answer

90

Worked Solutions

Arithmetic Sequences

Q233 MCQ

A sequence has first term 10, last term 70, and 13 terms. Select its sum:

Solution

$$1. S = \frac{13}{2}(10 + 70) = 520.$$

Correct option: B

Final answer

520

Q234 MCQ

A sequence has first term 2, last term 57, and 12 terms. Select its sum:

Solution

$$1. S = \frac{12}{2}(2 + 57) = 354.$$

Correct option: B

Final answer

354

Worked Solutions

Arithmetic Sequences

Q235 MCQ

A sequence has first term 25, last term 100, and 16 terms. Select its sum:

Solution

$$1. S = \frac{16}{2}(25 + 100) = 1000.$$

Correct option: B

Final answer

1000

Q236 MCQ

A sequence has first term 101, last term 200, and 100 terms. Select its sum:

Solution

$$1. S = \frac{100}{2}(101 + 200) = 15050.$$

Correct option: B

Final answer

15050

Worked Solutions

Arithmetic Sequences

Q237 MCQ

A sequence has first term 31, last term 91, and 16 terms. Select its sum:

Solution

1. $S = \frac{16}{2}(31 + 91) = 976.$

Correct option: B

Final answer

976

Q238 Written

Find the sum of the multiples of 3 among the two-digit natural numbers.

Solution

1. The sequence is 12, 15, 18, ..., 99, so $a = 12, d = 3, l = 99.$

2. $99 = 12 + (n - 1)3$ gives $n = 30.$

3. $S = \frac{30}{2}(12 + 99) = 1665.$

Final answer

1665

Worked Solutions

Arithmetic Sequences

Q239 **Written****Find the sum of all two-digit natural numbers that are multiples of 5.****Solution**

1. The sequence is 10, 15, ..., 95; $n = (95 - 10)/5 + 1 = 18$.
2. $S = \frac{18}{2}(10 + 95) = 945$.

Final answer

945

Q240 **Written****Find the sum of the multiples of 4 from 1 to 100.****Solution**

1. The sequence is 4, 8, ..., 100; $n = 25$.
2. $S = \frac{25}{2}(4 + 100) = 1300$.

Final answer

1300

Worked Solutions

Arithmetic Sequences

Q241 **Written****Find the sum of all multiples of 6 from 10 to 100.****Solution**

1. The sequence is 12, 18, ..., 96; $n = 15$.
2. $S = \frac{15}{2}(12 + 96) = 810$.

Final answer

810

Q242 **MCQ****Select the sum of the multiples of 3 from 3 to 99:****Solution**

1. $n = (99 - 3)/3 + 1 = 33$; $S = \frac{33}{2}(3 + 99) = 1683$.

Correct option: B**Final answer**

1683

Worked Solutions

Arithmetic Sequences

Q243 MCQ

Select the sum of the multiples of 4 from 4 to 100:**Solution**

$$1. \ n = (100 - 4)/4 + 1 = 25; S = \frac{25}{2}(4 + 100) = 1300.$$

Correct option: B**Final answer**

1300

Q244 Written

Find the sum of the natural numbers from 1 to 100 that leave a remainder of 1 when divided by 4.**Solution**

$$1. \ \text{The sequence is } 1, 5, \dots, 97; n = 25.$$

$$2. \ S = \frac{25}{2}(1 + 97) = 1225.$$

Final answer

1225

Worked Solutions

Arithmetic Sequences

Q245 **Written****Find the sum of all two-digit natural numbers that leave a remainder of 3 when divided by 5.****Solution**

1. The sequence is 13, 18, ..., 98; $n = (98 - 13)/5 + 1 = 18$.
2. $S = \frac{18}{2}(13 + 98) = 999$.

Final answer

999

Q246 **Written****Find the sum of all two-digit odd natural numbers.****Solution**

1. The sequence is 11, 13, ..., 99; $n = 45$.
2. $S = \frac{45}{2}(11 + 99) = 2475$.

Final answer

2475

Worked Solutions**Arithmetic Sequences****Q247** Written**Find the sum of all natural numbers from 10 to 100 that are not divisible by 6.****Solution**

1. Total 10 to 100 is 5005; multiples of 6 are 12 to 96 with sum 810.
2. Required sum = $5005 - 810 = 4195$.

Final answer

4195

Worked Solutions

Arithmetic Sequences

Q248 Challenge

Find the sum of the numbers from 1 to 100 which are not divisible by 4.

Solution

1. Total $S_{100} = 5050$. Multiples of 4 have sum 1300.
2. Required sum = $5050 - 1300 = 3750$.

Final answer

3750

Worked Solutions

Arithmetic Sequences

Q249 **Written**

In an arithmetic sequence with first term 4 and common difference 3, find how many terms from the first term have sum 50.

Solution

1. $50 = \frac{n}{2}\{2(4) + (n-1)3\} \Rightarrow 3n^2 + 5n - 100 = 0.$
2. The natural root is $n = 5$; $S_5 = 50.$

Final answer

5

Q250 **Written**

Find n when $a = 3, d = 2$ and $S_n = 63.$

Solution

1. $S_n = n(n+2) = 63 \Rightarrow n = 7.$

Final answer

7

Worked Solutions

Arithmetic Sequences

Q251 MCQ

Select the correct option: Find n if $S_n = 48$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 48 \Rightarrow n = 6$.

Correct option: B

Final answer

6

Q252 MCQ

Select the correct option: Find n if $S_n = 80$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 80 \Rightarrow n = 8$.

Correct option: B

Final answer

8

Worked Solutions

Arithmetic Sequences

Q253 MCQ

Select the correct option: Find n if $S_n = 120$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 120 \Rightarrow n = 10.$$

Correct option: B**Final answer**

10

Q254 MCQ

Select the correct option: Find n if $S_n = 168$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 168 \Rightarrow n = 12.$$

Correct option: B**Final answer**

12

Worked Solutions

Arithmetic Sequences

Q255 MCQ

Select the correct option: Find n if $S_n = 255$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 255 \Rightarrow n = 15.$

Correct option: B

Final answer

15

Q256 MCQ

Select the correct option: Find n if $S_n = 360$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 360 \Rightarrow n = 18.$

Correct option: B

Final answer

18

Worked Solutions

Arithmetic Sequences

Q257 MCQ

Select the correct option: Find n if $S_n = 440$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 440 \Rightarrow n = 20.$$

Correct option: B**Final answer**

20

Q258 MCQ

Select the correct option: Find n if $S_n = 624$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 624 \Rightarrow n = 24.$$

Correct option: B**Final answer**

24

Worked Solutions

Arithmetic Sequences

Q259 MCQ

Select the correct option: Find n if $S_n = 840$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 840 \Rightarrow n = 28.$

Correct option: B

Final answer

28

Q260 MCQ

Select the correct option: Find n if $S_n = 1088$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 1088 \Rightarrow n = 32.$

Correct option: B

Final answer

32

Worked Solutions

Arithmetic Sequences

Q261 MCQ

Select the correct option: Find n if $S_n = 1368$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 1368 \Rightarrow n = 36.$$

Correct option: B**Final answer**

36

Q262 MCQ

Select the correct option: Find n if $S_n = 1680$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 1680 \Rightarrow n = 40.$$

Correct option: B**Final answer**

40

Worked Solutions

Arithmetic Sequences

Q263 Written

For first term 17 and common difference -2 , find how many terms have sum -63 .

Solution

1. $S_n = n(18 - n)$.
2. $n(18 - n) = -63 \Rightarrow n^2 - 18n - 63 = 0$; hence $n = 21$.

Final answer

21

Q264 Written

Find n when $a = 24$, $d = -4$ and $S_n = -28$.

Solution

1. $S_n = n(26 - 2n) = -28 \Rightarrow n^2 - 13n - 14 = 0$; $n = 14$.

Final answer

14

Worked Solutions

Arithmetic Sequences

Q265 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 80$:**Solution**

1. $S_n = \frac{n}{2}(17 + 3n) = 80$; positive root $n = 5$.

Correct option: C**Final answer**

5

Q266 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 133$:**Solution**

1. $S_n = \frac{n}{2}(17 + 3n) = 133$; positive root $n = 7$.

Correct option: C**Final answer**

7

Worked Solutions

Arithmetic Sequences

Q267 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 198$:**Solution**

1. $S_n = \frac{n}{2}(17 + 3n) = 198$; positive root $n = 9$.

Correct option: C

Final answer

9

Q268 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 275$:**Solution**

1. $S_n = \frac{n}{2}(17 + 3n) = 275$; positive root $n = 11$.

Correct option: C

Final answer

11

Worked Solutions

Arithmetic Sequences

Q269 Written

Find the general term of $\{a_n\}$ with first term -60 and sum of the first 15 terms -60 .

Solution

1. $-60 = \frac{15}{2}\{2(-60) + 14d\} \Rightarrow d = 8.$
2. $a_n = -60 + 8(n - 1) = 8n - 68.$

Final answer

$$a_n = 8n - 68$$

Q270 Written

Find the general term with first term 8 and sum of the first 13 terms -130 .

Solution

1. $-130 = \frac{13}{2}(16 + 12d) \Rightarrow d = -3.$
2. $a_n = 8 - 3(n - 1) = 11 - 3n.$

Final answer

$$a_n = 11 - 3n$$

Worked Solutions

Arithmetic Sequences

Q271 **Written****Find the general term with first term 4 and $S_6 = 114$.****Solution**

$$1. \frac{6}{2}(8 + 5d) = 114 \Rightarrow d = 6.$$

$$2. a_n = 4 + 6(n - 1) = 6n - 2.$$

Final answer

$$a_n = 6n - 2$$

Q272 **Written****Find the general term when $a_3 = 10$ and $S_{10} = 175$.****Solution**

$$1. a + 2d = 10, 2a + 9d = 35 \Rightarrow d = 3, a = 4.$$

$$2. a_n = 4 + 3(n - 1) = 3n + 1.$$

Final answer

$$a_n = 3n + 1$$

Worked Solutions

Arithmetic Sequences

Q273 Written

For first term 65 and common difference -4 , find the number of terms giving the maximum sum and the sum.

Solution

1. $a_n = 65 - 4n$; $a_n > 0 \Rightarrow n < 17.25$.
2. The last positive term is $a_{17} = 1$; $S_{17} = \frac{17}{2}(65 + 1) = 561$.

Final answer

$$n = 17, S = 561$$

Q274 Written

For first term 70 and common difference -4 , find the number of terms giving the maximum sum and the sum.

Solution

1. $a_n = 70 - 4n$; $a_{18} = 2$ is the last positive term.
2. $S_{18} = \frac{18}{2}(70 + 2) = 648$.

Final answer

$$n = 18, S = 648$$

Worked Solutions

Arithmetic Sequences

Q275 Written

For $a = 20, d = -6$, find n giving the maximum sum and the sum.

Solution

1. $a_n = 26 - 6n$; the last positive term is $a_4 = 2$.
2. $S_4 = \frac{4}{2}(20 + 2) = 44$.

Final answer

$$n = 4, S = 44$$

Q276 Written

For $a = 126, d = -13$, find n giving the maximum sum and the sum.

Solution

1. $a_n = 139 - 13n$; the last positive term is $a_{10} = 9$.
2. $S_{10} = \frac{10}{2}(126 + 9) = 675$.

Final answer

$$n = 10, S = 675$$

Worked Solutions

Arithmetic Sequences

Q277 Written

Find the general term where $S_{10} = 10$ and $S_{20} = 40$.**Solution**

1. $2a + 9d = 2, 2a + 19d = 4 \Rightarrow d = \frac{1}{5}, a = \frac{1}{10}.$
2. $a_n = \frac{1}{10} + (n-1)\frac{1}{5} = \frac{2n-1}{10}.$

Final answer

$$a_n = \frac{2n-1}{10}$$

Q278 Written

Find the general term where $S_6 = 3$ and $S_{10} = 65$.**Solution**

1. $2a + 5d = 1, 2a + 9d = 13 \Rightarrow d = 3, a = -7.$
2. $a_n = -7 + 3(n-1) = 3n - 10.$

Final answer

$$a_n = 3n - 10$$

Worked Solutions

Arithmetic Sequences

Q279 **Written****Find the general term where $S_{20} = 400$ and $S_{30} = 900$.****Solution**

$$1. \ 2a + 19d = 40, \ 2a + 29d = 60 \Rightarrow d = 2, \ a = 1.$$

$$2. \ a_n = 1 + 2(n - 1) = 2n - 1.$$

Final answer

$$a_n = 2n - 1$$

Worked Solutions

Arithmetic Sequences

Q280 Challenge

The 15th term is 33 and the 50th term is 103. Find the sum of the first 79 terms.

Solution

1. $35d = 70 \Rightarrow d = 2, a = 5.$
2. $S_{79} = \frac{79}{2}\{2(5) + 78(2)\} = 6557.$

Final answer

6557

Worked Solutions

Arithmetic Sequences

Quiz MCQ 1 **Concept Quiz · MCQ**

For $a_n = \frac{2n+1}{n+2}$, what is a_5 ?:

Solution

1. Substitute $n = 5$: $a_5 = (2(5)+1)/(5+2) = 11/7$.

Correct option: A

Final answer

$$\frac{11}{7}$$

Quiz MCQ 2 **Concept Quiz · MCQ**

An arithmetic sequence has $a_3 = 10$ and $a_8 = 30$. Select its general term:

Solution

1. $10 = a + 2d$ and $30 = a + 7d$, so $d = 4$, $a = -2$.

2. $a_n = 4n - 2$.

Correct option: B

Final answer

$$a_n = 4n - 2$$

Worked Solutions

Arithmetic Sequences

Quiz MCQ 3 **Concept Quiz · MCQ**

For $a = 12$, $d = -2$, select the sum of odd-numbered terms through a_9 :

Solution

1. Odd terms are 12, 8, 4, 0, -4 ; their sum is 20.

Correct option: C

Final answer

20

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: If $S_{10} = 10$ and $S_{20} = 40$, find d :

Solution

1. $d = \frac{1}{5}$.

Correct option: B

Final answer

$\frac{1}{5}$

Worked Solutions

Arithmetic Sequences

Quiz WRITTEN 5 Concept Quiz · Written

If $a_n = 3n^2 - 2n + 1$, find a_1, a_2, a_3 .

Solution

1. Substitution gives $a_1=2$, $a_2=9$, and $a_3=22$.

Final answer

$$a_1 = 2, a_2 = 9, a_3 = 22$$

Quiz WRITTEN 6 Concept Quiz · Written

Find a_n for an arithmetic sequence with $a_4 = 14$ and $a_{10} = 62$.

Solution

1. $14 = a + 3d$ and $62 = a + 9d$, so $d = 8$, $a = -10$.

2. $a_n = 8n - 18$.

Final answer

$$a_n = 8n - 18$$

Worked Solutions

Arithmetic Sequences

Quiz WRITTEN 7 Concept Quiz · Written

For $a = 9, d = 4$, find the sum of odd-numbered terms from a_1 to a_{15} .

Solution

1. Selected terms have 8 terms, first 9, last 65.
2. $\text{Sum} = 8(9 + 65)/2 = 296$.

Final answer

296

Quiz WRITTEN 8 Concept Quiz · Written

Find the general term when $S_{20} = 400$ and $S_{30} = 900$.

Solution

1. $d = 2, a = 1$; therefore $a_n = 2n - 1$.

Final answer

$a_n = 2n - 1$

Answer key

Use the question number shown on each page.

Q001 $a_1 = -1, a_2 = 1, a_3 = 3$

Q002 $a_1 = -6, a_2 = -10, a_3 = -12$

Q003 $a_1 = -3, a_2 = 2, a_3 = 7$

Q004 $a_1 = -2, a_2 = -5, a_3 = -8$

Q005 $a_1 = -6, a_2 = -8, a_3 = -10$

Q006 $a_1 = 0, a_2 = 3, a_3 = 8$

Q007 $a_1 = 3, a_2 = 7, a_3 = 13$

Q008 $a_1 = 7, a_2 = 12, a_3 = 19$

Q009 B

Q010 C

Q011 D

Q012 A

Q013 B

Q014 C

Q015 D

Q016 A

Q017 B

Q018 C

Q019 D

Q020 A

Q021 $a_1 = 6, a_2 = 18, a_3 = 54$

Q022 $a_1 = 2, a_2 = 8, a_3 = 26$

Q023 $a_1 = -3, a_2 = 9, a_3 = -27$

Q024 $a_1 = -6, a_2 = 12, a_3 = -24$

Q025 A

Q026 B

Q027 C

Q028 D

Q029 A

Q030 B

Q031 C

Q032 D

Q033 $a_1 = \frac{1}{2}, a_2 = \frac{2}{3}, a_3 = \frac{3}{4}$

Q034 $a_1 = \frac{1}{3}, a_2 = \frac{2}{5}, a_3 = \frac{3}{7}$

Q035 $a_1 = 1, a_2 = \frac{1}{4}, a_3 = \frac{1}{7}$

Q036 $a_1 = \frac{1}{2}, a_2 = \frac{4}{5}, a_3 = 1$

Q037 $a_1 = 2, a_2 = 1, a_3 = \frac{4}{5}$

Q038 C

Q039 D

Q040 A

Q041 B

Q042 C

Q043 D

Q044 $a_n = (n-1)^2$

Q045 $a_n = (n-1)^3$

Q046 $a_n = n^2$

Q047 $a_n = (n+6)^2$

Q048 B

Q049 C

Q050 D

Q051 A

Q052 B

Q053 C

Q054 $a_n = 3n - 1$

Q055 $a_n = 2n - 2$

Q056 $a_n = 3n - 3$

Q057 $a_n = 5n - 1$

Q058 $a_n = 7n - 1$

Q059 A

Q060 B

Q061 C

Q062 D

Q063 $a_n = (-1)^{n-1} 2n$

Q064 $a_n = (-1)^n 3n$

Q065 $a_n = (-1)^n 5n$

Q066 $a_n = (-1)^n (3n - 2)$

Q067 $a_n = \frac{n}{2n-1}$

Q068 $a_n = \frac{2n-1}{n^2}$

Q069 $a_n = \frac{(-1)^{n-1}}{2n-1}$

Q070 $a_n = \frac{n}{n+1}$

Q071 $a_n = n 3^n$

Q072 $a_n = \frac{n}{n+1}; a_{20} = \frac{20}{21}$

Q073 (a) $a = 2, d = 3(b) a = 3, d = -5$

Q074 (a) $a = 1, d = 4(b) a = 23, d = -4$

Q075 B

Q076 A

Q077 C

Q078 D

Q079 (a)(3, 2); (b)(-13, 4); (c)(-2, 3); (d)(10, -3); (e)(6, $-\frac{2}{3}$); (f)(-4, $-\frac{1}{2}$).

Q080 B

Q081 A

Q082 C

Q083 B

Q084 (a) $d = -3, A = 5, B = -4(b) d = -5, A = 8, B = 3$

Q085 (a) $d = 7, A = 16, B = 23(b) d = -6, A = 23, B = 11$

Q086 A

Q087 C

Q088 B

Q089 (a)(d) = 7, A = 22, B = 20; (b)(d) = 6, A = 19, B = 21; (c)(d) = -5, A = 14, B = -1; (d)(d) = -2, A = 17, B = 13; (e)(d) = 3, A = 14, B = 17; (f)(d) = -4, A = 13, B = 9, C = 1.

Q090 (a) $a_n = 3n + 1, a_7 = 22(b) a_n = 5 - 3n, a_7 = -16$

Q091 (a) $a_n = 4n + 1, a_7 = 29$; (b) $a_n = 3n - 5, a_7 = 16$; (c) $a_n = 13 - 3n, a_7 = -8$; (d) $a_n = -\frac{n}{2}, a_7 = -\frac{7}{2}$.

Q092 $a_n = 8 + (n-1)(-6) = 14 - 6n; a_{20} = -106$

Q093 (a) $a_n = 3n - 10, a_{10} = 20$; (b) $a_n = 1 - 3n, a_{10} = -29$; (c) $a_n = \frac{n+13}{2}, a_{10} = \frac{23}{2}$; (d) $a_n = \frac{3-n}{3}, a_{10} = -\frac{7}{3}$.

Q094 $a = 7, d = 3$.

Q095 $a_n = 5n - 10$

Q096 $a_n = 4n + 2$

Q097 $a_n = 23 - 3n$

Q098 $a_n = 9n - 17$

Q099 $a_n = -5n - 13$

Q100 D

Q101 B	Q133 A	Q165 144	Q197 C
Q102 A	Q134 B	Q166 -165	Q198 C
Q103 B	Q135 $n = 125$, the 125th term	Q167 176	Q199 $2n^2 - n$
Q104 $a_n = 3n + 33$	Q136 $n = 35$, the 35th term	Q168 360	Q200 $\frac{n(7-3n)}{2}$
Q105 $a_n = 2n - 12$	Q137 A	Q169 C	Q201 $\frac{5n(n-1)}{2}$
Q106 $a_n = 13 - 4n$	Q138 D	Q170 B	Q202 $5n - 2n^2$
Q107 $a_n = 18 - 7n$	Q139 D	Q171 C	Q203 $-n^2 - 3n$
Q108 A	Q140 B	Q172 B	Q204 414
Q109 D	Q141 $n = 72$, the 72nd term	Q173 C	Q205 5050
Q110 C	Q142 $n = 171$, the 171st term	Q174 B	Q206 20100
Q111 D	Q143 B	Q175 392	Q207 11325
Q112 $a_n = 8n - 14$	Q144 B	Q176 343	Q208 2485
Q113 $a_n = 20 - 3n$	Q145 C	Q177 741	Q209 5050
Q114 $a_n = 8n - 18$	Q146 A	Q178 354	Q210 80200
Q115 $a_n = 42 - 4n$	Q147 B	Q179 1716	Q211 31375
Q116 $a_n = 79 - 3n$	Q148 $a = 4$	Q180 C	Q212 8085
Q117 $a_n = 8n - 18$	Q149 $x = -4$ or $x = 3$	Q181 C	Q213 24535
Q118 C	Q150 $x = -2$ or $x = 4$	Q182 C	Q214 4950
Q119 C	Q151 $x = -1$ or $x = 3$	Q183 C	Q215 B
Q120 B	Q152 $x = 0$, $y = -2$, $z = -4$	Q184 C	Q216 B
Q121 A	Q153 105	Q185 B	Q217 B
Q122 A	Q154 100	Q186 0	Q218 B
Q123 B	Q155 120	Q187 208	Q219 B
Q124 $a_n = \frac{2n-37}{5}$	Q156 152	Q188 B	Q220 B
Q125 A	Q157 155	Q189 C	Q221 B
Q126 D	Q158 B	Q190 C	Q222 B
Q127 $n = 28$, the 28th term	Q159 B	Q191 C	Q223 B
Q128 $n = 14$, the 14th term	Q160 B	Q192 B	Q224 B
Q129 $n = 22$, the 22nd term	Q161 B	Q193 C	Q225 15050
Q130 $n = 40$, the 40th term	Q162 B	Q194 66	Q226 25050
Q131 C	Q163 C	Q195 750	Q227 4905
Q132 D	Q164 -130	Q196 154	

Q228 B	Q244 1225	Q260 B	Q276 $n = 10, S = 675$
Q229 B	Q245 999	Q261 B	Q277 $a_n = \frac{2n-1}{10}$
Q230 B	Q246 2475	Q262 B	Q278 $a_n = 3n - 10$
Q231 B	Q247 4195	Q263 21	Q279 $a_n = 2n - 1$
Q232 B	Q248 3750	Q264 14	Q280 6557
Q233 B	Q249 5	Q265 C	Quiz MCQ 1 A
Q234 B	Q250 7	Q266 C	Quiz MCQ 2 B
Q235 B	Q251 B	Q267 C	Quiz MCQ 3 C
Q236 B	Q252 B	Q268 C	Quiz MCQ 4 B
Q237 B	Q253 B	Q269 $a_n = 8n - 68$	Quiz WRITTEN 5 $a_1 = 2, a_2 = 9, a_3 = 22$
Q238 1665	Q254 B	Q270 $a_n = 11 - 3n$	Quiz WRITTEN 6 $a_n = 8n - 18$
Q239 945	Q255 B	Q271 $a_n = 6n - 2$	Quiz WRITTEN 7 296
Q240 1300	Q256 B	Q272 $a_n = 3n + 1$	Quiz WRITTEN 8 $a_n = 2n - 1$
Q241 810	Q257 B	Q273 $n = 17, S = 561$	
Q242 B	Q258 B	Q274 $n = 18, S = 648$	
Q243 B	Q259 B	Q275 $n = 4, S = 44$	